

Full Stack Developer – Technical Assessment

Objective

The purpose of this assessment is to evaluate your ability to design and develop a full-stack web application with authentication, product management, cart system, and role-based access control.

You are expected to implement both **client-side and server-side features**, along with proper **API design, database structure, and authentication flow**.



Deadline for Submission:

Your deadline for completing this assessment is **5 days from the date of assignment**. Please ensure the project is completed and submitted within the given timeline.

Key Objectives

- Implement secure user authentication using **Email OTP**
 - Build a **cart system** supporting both guest and logged-in users
 - Develop a **product management system with variations**
 - Create a **role-based access system (User & Admin)**
 - Design clean and structured **REST APIs**
 - Maintain proper **database schema & state management**
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Client-Side Features

1. Authentication System

Implement a complete authentication flow:

- User Registration (Name, Email, Password)
- Login with Email & Password
- Email OTP Verification (during registration / login / reset)
- Forgot Password & Reset Password flow
- Session handling (JWT / Cookies)

OTP can be implemented using:

- Mock service OR
 - Real email service (**Bonus**)
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2. User Profile Management

- View profile details
 - Edit profile information (Name, Email, etc.)
 - Persist updated data in database
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3. Cart Management

Guest Cart

- Add products without login
- Store cart locally (localStorage / sessionStorage)

Logged-in Cart

- Sync guest cart after login
- Store cart in database
- Update quantity / remove items

Important:

Proper **merging logic between guest cart and user cart** must be implemented.

4. Product Listing

Display products with:

- Product Name
 - Product Image
 - Price
 - Available variations (Size, Color, etc.)
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5. Product Details Page

Show:

- Product description
 - Variation selection (size / color)
 - Add to cart button
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Admin Panel Features

1. Admin Authentication

- Secure admin login
 - Role-based access control
 - Only admin can access dashboard
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2. Product Management

Admin should be able to:

- Create products
 - Edit products
 - Delete products
 - Upload product images
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3. Category Management

- Create categories
 - Assign categories to products
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4. Product Variations

Implement flexible product variation system:

Examples:

- Size (S, M, L, etc.)
- Color (Red, Blue, etc.)

Admin should be able to:

- Add variations
 - Edit variations
 - Remove variations per product
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API Requirements

Candidate should design REST APIs similar to below:

Auth APIs

- POST /register
- POST /login
- POST /send-otp
- POST /verify-otp
- POST /forgot-password
- POST /reset-password

User APIs

- GET /profile
- PUT /profile

Product APIs

- GET /products
- GET /products/:id

Cart APIs

- POST /cart
- GET /cart
- PUT /cart
- DELETE /cart/item

Admin APIs

- POST /admin/login
 - POST /products
 - PUT /products/:id
 - DELETE /products/:id
 - POST /categories
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Recommended Tech Stack

- Frontend: React.js / Next.js (Preferred)
- Backend: Node.js + Express
- Database: MongoDB / MySQL
- Authentication: JWT / Sessions
- Styling: Tailwind CSS